Fibula : From AI Extraction to Autonomous Workflows

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Manual reconciliation across invoices, purchase orders, GRNs, BOMs, and work orders is typically slow, fragmented, and error-prone. This project began by analyzing Accounts Payable (AP) and Bill of Materials (BOM) reconciliation logic—specifically evaluating key documents, extraction fields, matching rules, and variance checks. Building on this process analysis, we conducted over 15 user tests and workflow evaluations on Fibula, Staple AI's node-based workflow automation platform. Through this research, we identified critical usability and functionality gaps, allowing us to propose targeted improvements that help users transition seamlessly from basic AI extraction to trusted, autonomous reconciliation workflows.

Introduction

Industrial document reconciliation suffers from financial leakage due to legacy, stateless extraction engines. Fibula resolves this by allowing operators to describe requirements in plain natural language, prompting autonomous AI agents to generate stateful, node-based workflows. This empowers non-technical users to independently build, troubleshoot, and launch complex reconciliation pipelines, driving manual support dependencies down to zero.

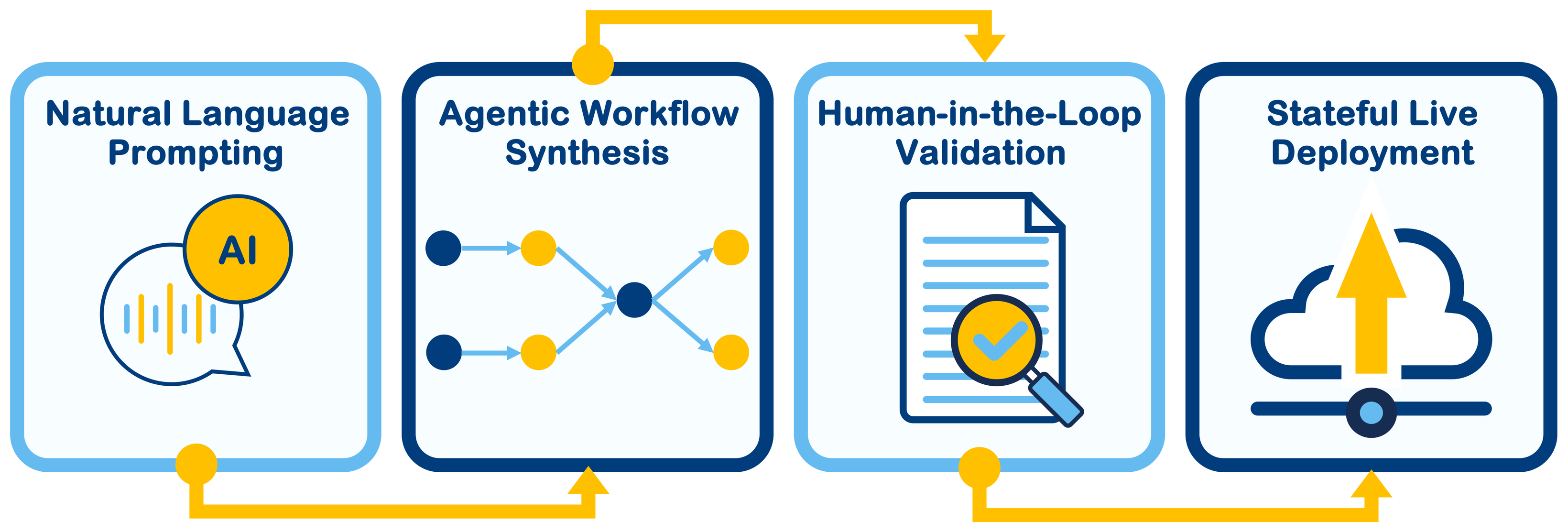

Figure 1: Agentic Workflow Synthesis from Natural Language Prompting to Stateful Live Deployment

Methodology

Reconciliation workflow analysis

Mapped AP/BOM reconciliation logic, including document inputs, extraction fields, matching keys, variance checks, and exception outcomes.

Task based user testing

Observed users completing Fibula workflow tasks through semi-structured interviews and think-aloud testing.

1. The 90% "No-Help" Rule: Researchers stayed passive; major blockers were recorded as system usability failures.
2. Evidence-Based Coding: Design recommendations were based on observed user friction, not personal preference.

User Journey & Mental Model Analysis

Mapped the end-to-end experience from workflow creation to output validation, then compared what users expected the AI to do with what Fibula actually required them to configure, test, and interpret.

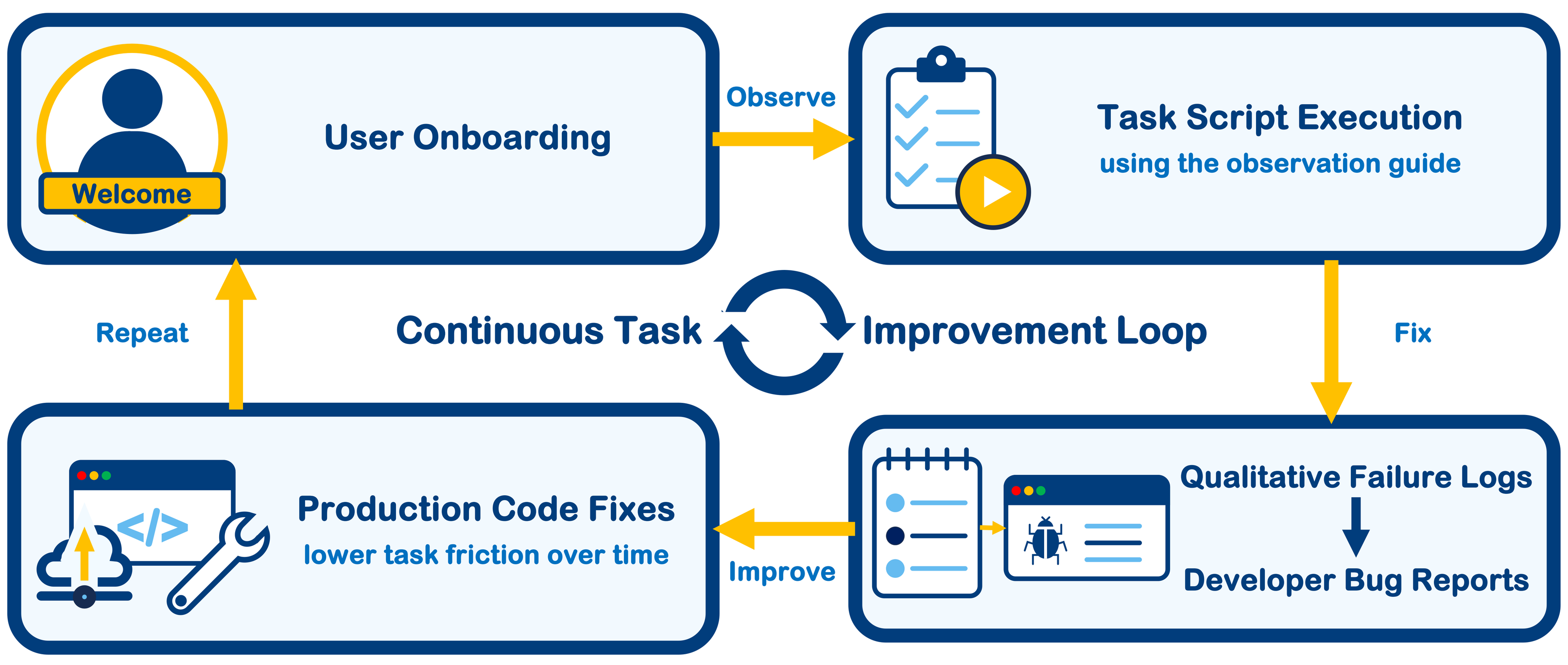

Figure 2: Evidence-Based Design and Validation Sprint Loop Framework

Conclusion

Fibula shows strong potential to move industrial reconciliation from AI extraction to autonomous workflow automation. To improve adoption, the platform should provide clearer workflow guidance, visible progress feedback, and actionable reconciliation outputs such as matched/unmatched results, variance reasons, and next-step recommendations.

Key Findings

Spatial Desynchronization (Mind-Map vs. Black Box)

- Silent backend processing destroys system trust; industrial workflows require immediate visual confirmation to maintain behavioral continuity.
- Highly surprised by the high density of structural keyword extraction; explicitly praised the node diagram layouts, comparing the visual logic to an intuitive "mind map."
- Engineered WebSocket-driven live canvas animation that dynamically draws, links, and updates node component states in real time as the AI Agent generates the workflow.

Convenience Bias & Cognitive Choice Paralysis

- High information density on compact displays causes fatigue, forcing users to select imperfect default options to avoid the friction of custom text typing.
- Hick's Law and the Principle of Least Effort dictate that presenting verbose summaries and excessive action options prompts users to sacrifice intent accuracy for immediate interaction speed.
- Restricted conversational suggestions to 2–3 predictive intent choice chips + a fallback choice, supported by fluid, draggable table-panel expansion rules.

Operational Opacity & Interaction Dead-Ends

- Silent background integration pauses and obscure system errors act as terminal user boundaries, triggering immediate session abandonment.
- Strongly satisfied with core parsing cleanliness and data alignment when uploading standard unstructured document types (PDFs and Images).
- Deprecated silent processing states in favor of Natural-Language Advice Cards, automated pre-ingestion review summaries, and in-modal workflow rerun shortcuts.

Analysis

Journey level friction

Problems appeared across the workflow, from setup and upload to testing, output review, and recovery.

Mental Model gap

Users expected AI to guide and validate the workflow, but still had to understand templates, nodes, fields, and rules.

Reconciliation risks

Unclear setup and weak feedback reduce trust in matched/unmatched results and variance decisions.

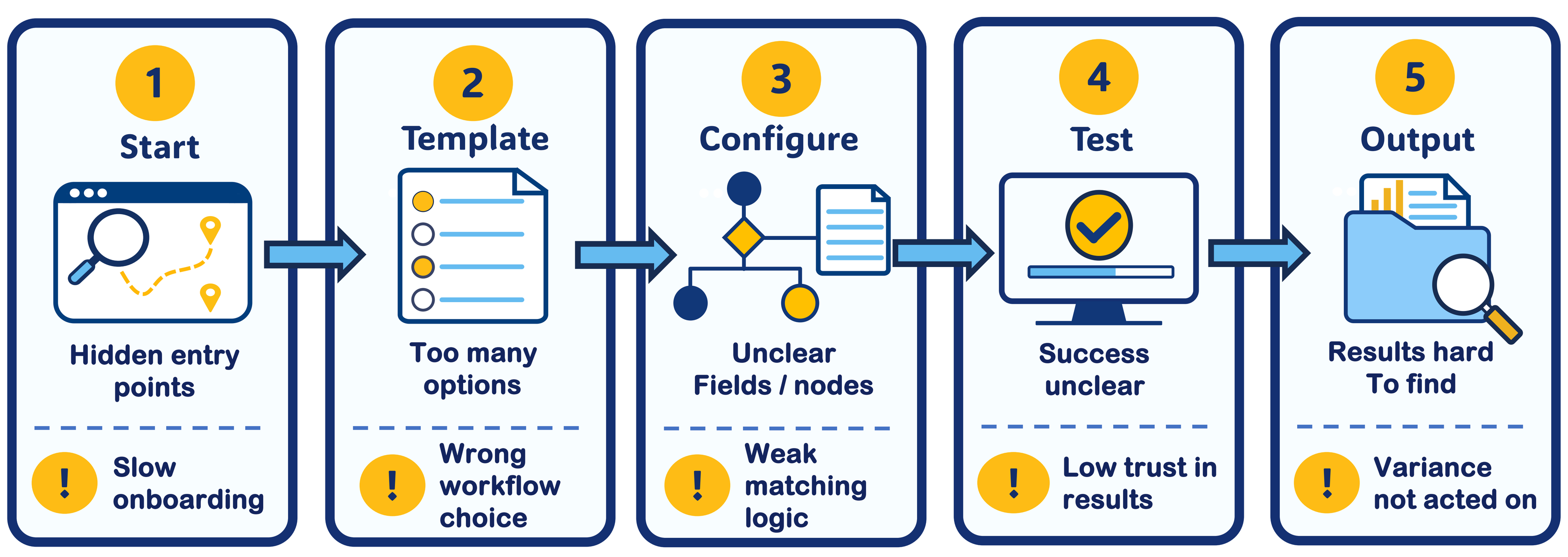

Figure 3: Journey to risk map

Scan to explore

Acknowledgements

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